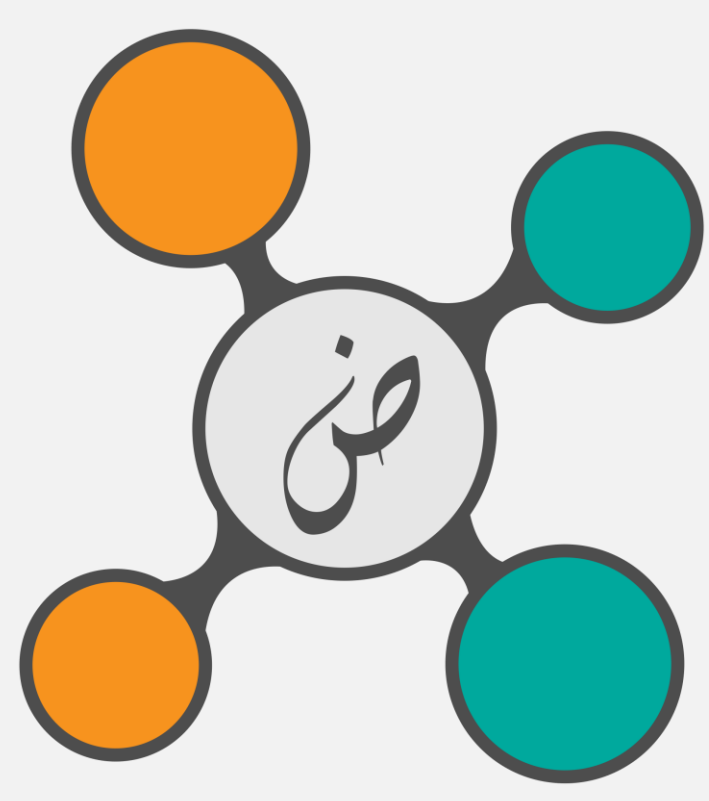




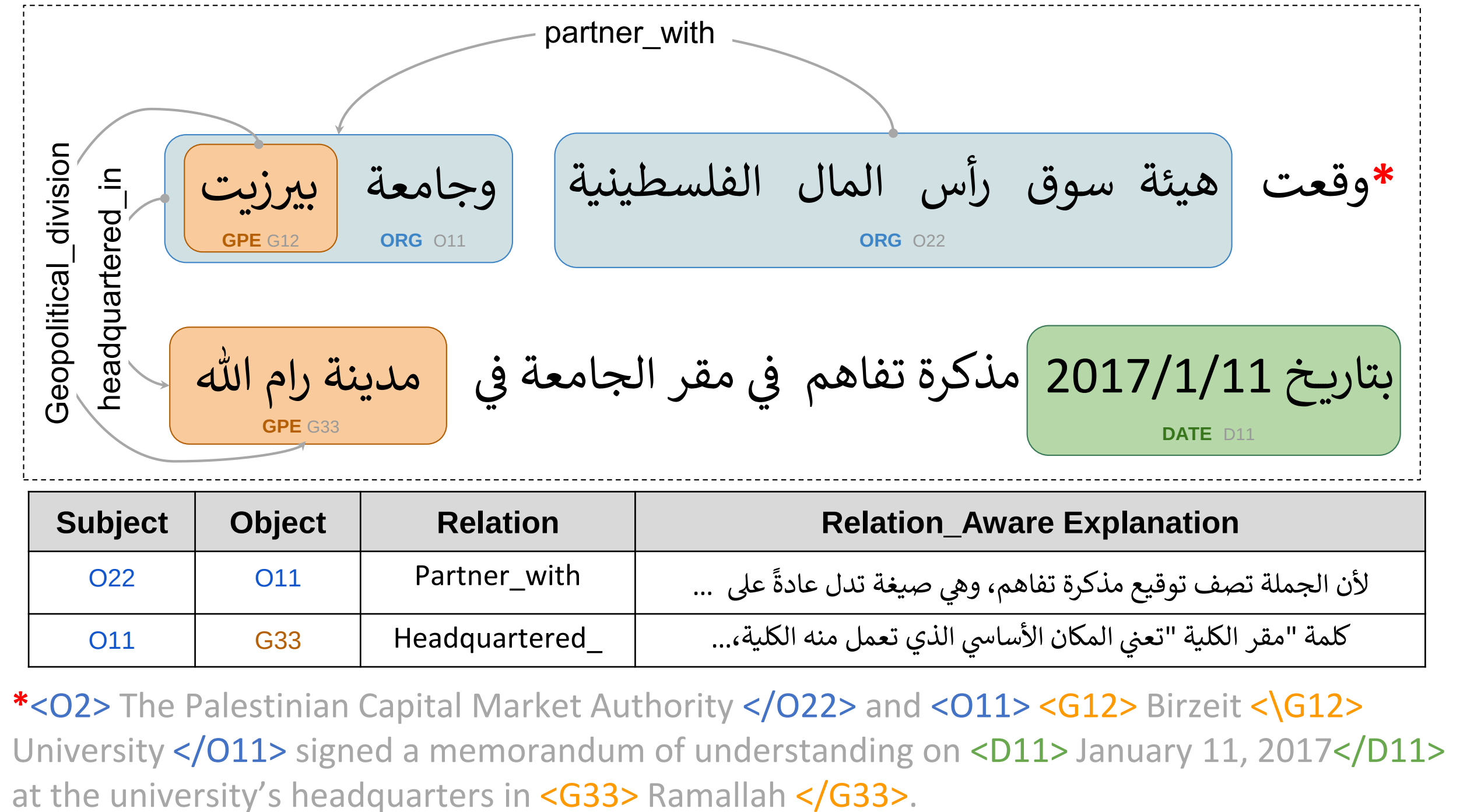
Wojood^{Relations}: Arabic Relation Extraction Corpus and Modeling

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Summary

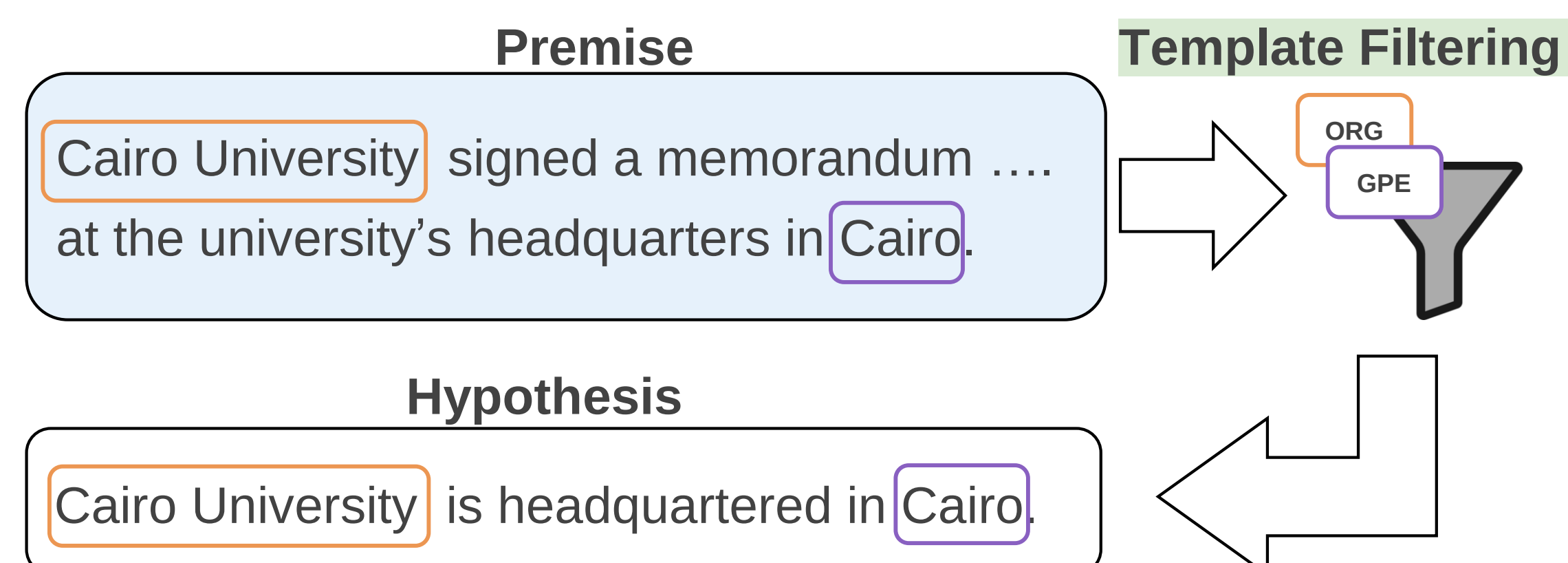
- The largest Arabic relation extraction corpus.
- 550K tokens, 33K sentences, covering multiple domains.
- Manually annotated with **40 relation types**, ~15K triples.
- Supports **21 entity types**, including nested entities.
- Inner annotator agreement **Cohen's k of 0.92**.
- Open source (CC BY-4.0).



Arabic Relation Extraction Modeling

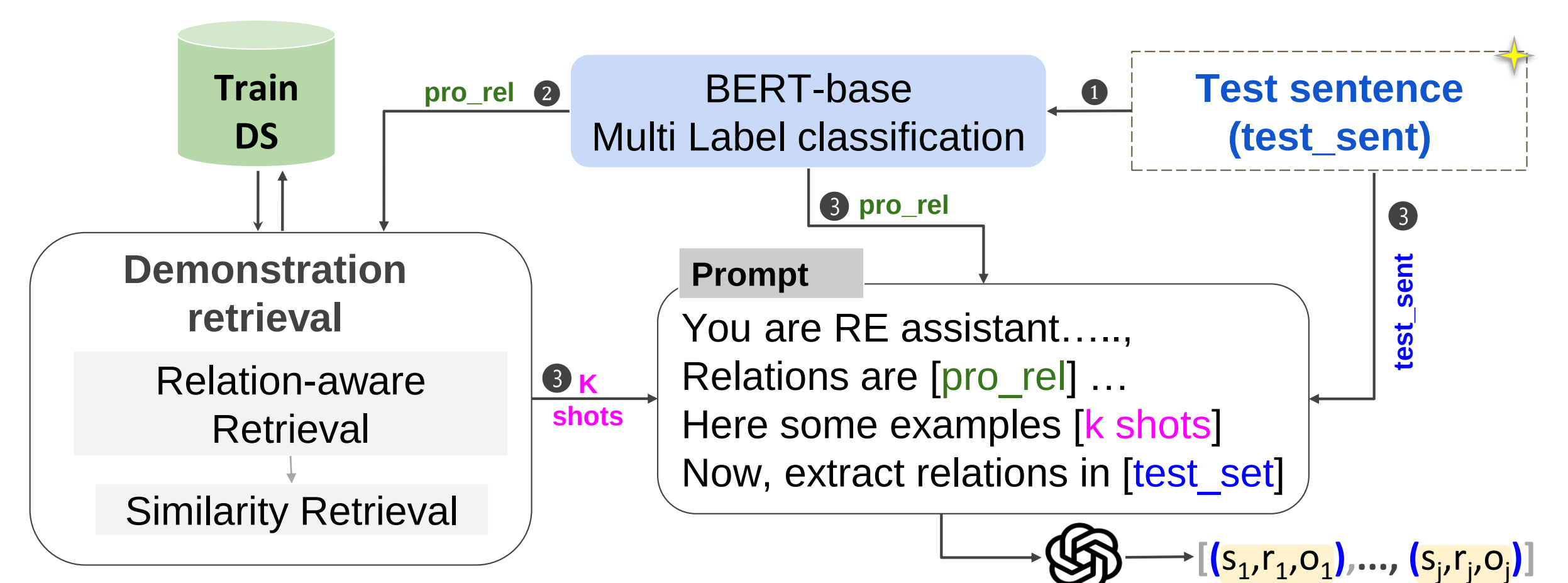
NLI-RE, frames relation extraction as a **binary Natural Language Inference (NLI)** problem with **relation-aware templates**.

- 1) **Relation Templates**: Define a relation-aware template for each relation type, constrained by domain and range types.
- 2) **Premise–Hypothesis Generation**: Treat the sentence as the premise and generate hypotheses for each entity pair using matching templates.
- 3) **Labeling**: Assign True if the relation exists in WojoodRelations, otherwise False.
- 4) **Classification**: Encode premise–hypothesis pairs with BERT and classify relations.



GPT-Joint, LLM-based **joint entity and relation extraction**.

- 1) **Relation Type Prediction**: Use a fine-tuned multi-label BERT to predict candidate relation types with confidence above a threshold.
- 2) **Relation-Aware Retrieval**: Retrieve top- k semantically similar examples sharing predicted relations for in-context use.
- 3) **LLM Prompting (Few-shot)**: Construct a prompt with task instructions, predicted relations, and top- k retrieved examples enriched with rationales.



Baselines using Wojood^{Relations}

Relation Extraction Baselines:

- RIFRE: Supervised graph-based method.
- GPT-RE: LLM-based method.
- NLI-RE: Supervised NLI-based method.

Joint Entity and RE Baselines:

- GPT-Joint: LLM-based joint entity and relation extraction.

Conclusion

Wojood^{Relations}: largest Arabic RE corpus (40 relation types). Supervised models outperform LLMs.

Remaining challenges: implicit and long-tail relation detection.

Future work: better generalization & improve contextual reasoning.

Relation Extraction Baselines Results:

Model	Micro P	Micro R	Micro F1	Macro P	Macro R	Macro F1
NLI-RE (supervised)	88.91	88.65	88.61	89.00	88.50	88.58
RIFRE (supervised)	93.29	93.28	92.89	54.36	54.34	52.05
GPT-RE (LLM)	89.25	83.66	85.78	69.20	68.75	63.55

Joint Entity and RE Baselines Results:

Model	Zero-Shot			Few-Shot		
	P	R	F1	P	R	F1
Llama3-70b	35.00	28.43	31.37	59.06	67.58	63.03
deepseek-distill-llama	27.24	18.50	22.03	74.45	61.16	67.15
gpt-4o (with LLM relation proposal)	43.30	29.83	35.32	80.39	60.20	68.85
gpt-4o (with BERT relation proposal)	-	-	-	65.92	62.90	64.38
deepseek-reasoner	35.35	29.02	31.87	76.20	69.56	72.73



Paper



(Data, Code, Demo)



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Downloads and Demo

<https://sina.birzeit.edu/relations/>